

MATH PUZZLES

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1 Problems

Problem 1. Without solving for x , show that if

$$x + \frac{1}{x} = 1, \text{ then } x^7 + \frac{1}{x^7} = 1.$$

Problem 2. Find the smallest positive integer divisible by 15 which has exactly 15 divisors.

Problem 3. Show that there is an element in the infinite sequence

$$7, 77, 777, 7777, \dots$$

that is divisible by 2003.

Problem 4 (Harder). The set A consists of $n + 1$ positive integers, none of which have a prime divisor greater than the n th smallest prime. Show that there exists a nonempty subset $B \subseteq A$ where the product of the elements of B is a perfect square.

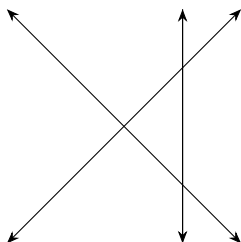
An integer is a perfect square if it is the square of another integer. I.e., the set of perfect squares is $\{n^2 \mid n \in \mathbb{N}\}$.

Problem 5. You have an n -sided die. What is the expected number of rolls it takes to land on each face at least once?

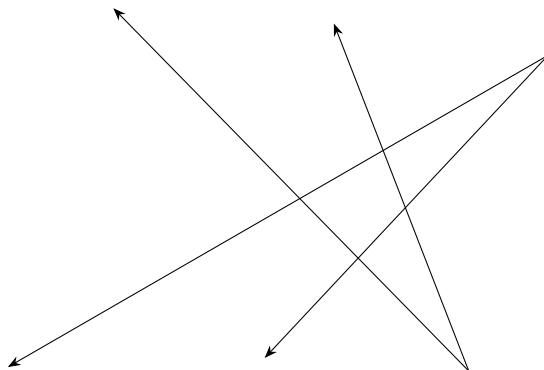
Problem 6. There are 100 light switches, all originally off. One hundred people pass by the switches; the first person flips every one, the second flips every other one, the third every third one, and so on. After the 100th person passes, which lights are on?

Problem 7. There are 100 noodles. Uniformly at random, you select two noodle ends and connect them, repeating the process until there are no more ends to connect. On average, how many loops form, and what is the probability that just one loop forms?

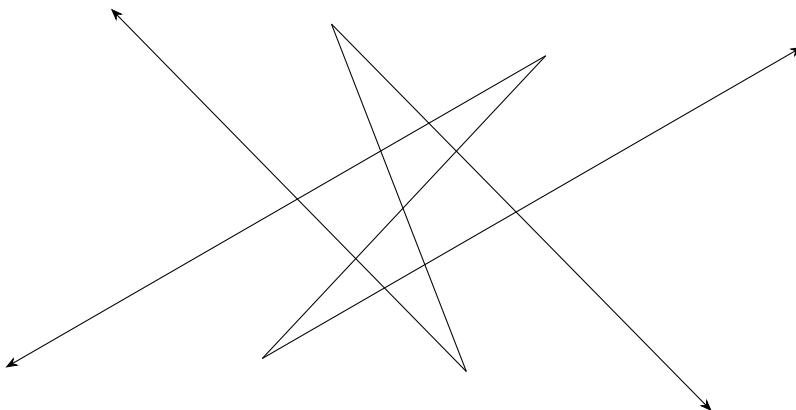
Problem 8. What is the maximum number of regions n lines segment the 2D plane into? $L_0 = 1$, $L_1 = 2$, $L_2 = 4$, $L_3 = 7$.



Problem 9. What is the maximum number of regions n zigs segment the 2D plane into? Each zig is made up of a pair of half-infinite lines that meet at a point. $Z_2 = 7$.



Problem 10. What is the maximum number of regions that n zig-zag lines can segment the 2D plane into? Each zig-zag is made up of a pair of half-infinite parallel lines and a finite one connecting them. $ZZ_2 = 12$.



2 Solutions

Solution 1. Consider the powers of x :

$$\begin{aligned}x^2 &= x - 1; \\x^3 &= x^2 - x \\&= x - 1 - x \\&= -1.\end{aligned}$$

Therefore $x^6 = (x^3)^2 = (-1)^2 = 1$, so $x^7 = x$.

Solution 2. By the fundamental theorem of algebra, every positive integer can be written as a product of prime powers, $\prod_{i=1}^k p_i^{r_i}$.

It's not too hard to reason about the number of divisors of a product of powers. For each prime power we can use anywhere between 0 and all of the powers of the prime as a divisor. E.g., if our number is $81 = 3^4$ then we can choose zero through four "multiplicative copies" of 3 as divisors of 81. This count is independent across each prime, so

$$\# \text{ of divisors of } \prod_{i=1}^k p_i^{r_i} = \prod_{i=1}^k r_i + 1.$$

The factors of 15 are only 3 and 5, so the least number which has 15 divisors and divides 15 must have a prime factorization of the form $3^a 5^b$, where $a, b \geq 1$. However, the product of $a + 1$ and $b + 1$ *must also equal* 15 to give 15 divisors. I.e.,

$$\begin{aligned}(a + 1)(b + 1) &= 3 \times 5 \\ \implies (a = 2 \text{ and } b = 4) \text{ or } (a = 4 \text{ and } b = 2).\end{aligned}$$

And since we want $3^a 5^b$ to be as small as possible, we put the larger power on the smaller prime, giving our final answer: $3^4 5^2 = 2025$.

Solution 3. Let a_i denote the i th element of the sequence so $a_1 = 7$, $a_2 = 77$, and so on. There are an infinite number of elements in our sequence but only a finite number of remainders when dividing any element in the sequence by 2003. So, by the pigeon hole principle, there must exist two elements $a_i \neq a_j$ which have the same remainder. That is, for $m, n \in \mathbb{N}$,

$$\begin{aligned}a_i &= m \times 2003 + r \\ a_j &= n \times 2003 + r.\end{aligned}$$

Without loss of generality, suppose $a_j > a_i$. Then $a_j - a_i = (m - n) \times 2003$, so 2003 divides $a_j - a_i$.

But $a_j - a_i$ is not an element of the sequence, so let's look at it more closely using an example. Consider $a_9 - a_5$:

$$\begin{aligned} a_9 - a_5 &= \begin{array}{r} 777777777 \\ - \quad 77777 \\ \hline 777700000 \end{array} \\ &= a_4 \times 10^5. \end{aligned}$$

So in general $a_j - a_i = a_{j-i} \times 10^i$.

Since 2003 divides $a_{j-i} \times 10^i$, but 2003 shares no factors with any power of 10 (neither 2 or 5 divide 2003), 2003 must divide a_{j-i} .¹

Solution 4.

Notation. Let the product of the elements of a set S be denoted by $\Pi(S)$. I.e.,

$$\Pi(S) := \prod_{s \in S} s.$$

Given a prime factorization $m = \prod_{i=1}^n p_i^{e_i}$, where p_i denotes the i th prime, let the “parity signature” of m be the bit string which records the parity of each prime exponent of m , denoted by $\text{ps}(m)$. I.e., using the notation of formal language theory with alphabet $\Sigma = \{0, 1\}$,

$$\text{ps}(m) := \prod_{i=1}^n \text{parity } e_i,$$

where

$$\text{parity } x := \begin{cases} 0 & \text{if } x \text{ is even} \\ 1 & \text{if } x \text{ is odd.} \end{cases}$$

For example (if $n = 5$), $\text{ps}(2^3 11^{71}) = 2^3 3^0 5^0 7^0 11^{71} = 10001$.

Proof. “None of which have a prime divisor greater than the n th smallest prime” is the same as “every element of A has only the first n primes in its prime factorization,” so each $a \in A$ can be written as $\prod_{i=1}^n p_i^{e_i}$.

We want to show that there exists a nonempty subset $B \subseteq A$ such that $\text{ps}(\Pi(B)) = 0^n$.² To do this, we first want to show that there exist two nonempty subsets $C, D \subseteq A$ whose element products have the same parity signature.

$|A| = n + 1$, so there are $2^{n+1} - 1$ nonempty subsets of A . Let

$$P = \{\text{ps}(\Pi(B)) \mid B \subseteq A \text{ and } B \text{ nonempty}\}.$$

¹I.e., we use the fact that if c divides ab and $\gcd(c, b) = 1$, then c divides a .

²An integer is a perfect square if and only if its parity signature is an all-zero string because if each prime exponent is even, we can factor out a 2 from each of them, and since $x^{ca} y^{cb} = (x^a y^b)^c$, $\prod_{i=1}^n p_i^{2f_i} = (\prod_{i=1}^n p_i^{f_i})^2$.

Then $|P| = 2^{n+1} - 1$ only if the parity signatures of each $\Pi(B)$ are distinct. But there are only 2^n unique bit strings of length n , and

$$2^{n+1} - 1 > 2^n \iff 2^n > 1 \quad \forall n \geq 1.$$

So, by the pigeon hole principle, there exist distinct $C, D \subseteq A$ whose element products have the same parity signature.³ And if

$$\text{ps}(\Pi(C)) = \text{ps}(\Pi(D)),$$

then

$$\text{ps}(\Pi(C) \times \Pi(D)) = 0^n \tag{1}$$

because powers are added when numbers are multiplied, and an odd plus and odd is even, and an even plus an even is even.⁴

However,

$$\Pi(C) \times \Pi(D) \neq \Pi(C \cup D) \tag{2}$$

if $C \cap D \neq \emptyset$ since each member of $C \cap D$ appears twice in the product on the LHS of (2) but only once in the product on the RHS. But it's not too hard to show that if we remove the doubled members of the product from $C \cap D$ we will still be left with a perfect square. That is, we can quickly show that if ab is a perfect square and b is a perfect square, then a is a perfect square. By definition, if ab and b are perfect squares, then $\exists m, n \in \mathbb{Z} : m^2 = ab$ and $n^2 = b$. So $\frac{m^2}{n^2} = a$ and thus $(\frac{m}{n})^2 = a$.

Recall from (1) that $\Pi(C) \times \Pi(D)$ is a perfect square. So

$$\begin{aligned} \Pi(C) \times \Pi(D) &= \Pi(C \setminus D) \times \left(\prod_{i \in C \cap D} i \right)^2 \times \Pi(D \setminus C) \\ &= \Pi(C \Delta D) \times \left(\prod_{i \in C \cap D} i \right)^2. \end{aligned}$$

Therefore $\Pi(C \Delta D)$ is a perfect square, and $C \Delta D \subseteq A$ is nonempty since we only considered nonempty subsets of B when we originally applied the pigeon hole principle.⁵ $\square$

Alternate proof of Problem 4. Since the members of A are positive and have no divisor greater than the n th prime, we can write each $a \in A$ as

$$a = \prod_{i=1}^n p_i^{e_i(a)},$$

³In other words, there are more nonempty subsets of A than possible parity signatures, so at least two (distinct) nonempty subsets must produce the same parity signature.

⁴Any odd plus another odd is even because $(2m+1) + (2n+1) = 2(m+n+1)$.

⁵ Δ denotes the *symmetric difference* of two sets: $A \Delta B := (A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B)$.

where p_i is the i th prime and $e_i(a) \geq 0$ is the exponent of p_i in the prime factorization of a .

Now define a map $f : A \rightarrow \mathbb{F}_2^n$ such that

$$f(x) = (e_1(x) \bmod 2, e_2(x) \bmod 2, \dots, e_n(x) \bmod 2).$$

Applying f records the parity of the exponents of the primes in the prime factorization of x , and if all said parities are zero, then x is a perfect square.⁶

Now arbitrarily order the elements of A such that $\{a_1, a_2, \dots, a_{n+1}\} = A$, and apply f to each $a \in A$ to produce $n+1$ n -dimensional vectors, $v_1, v_2, \dots, v_{n+1}$, in $\mathbb{F}_2^n$. Since there are more vectors than the dimension of the vector space, n , $\{v_1, \dots, v_j\}$ must be linearly dependent, i.e., there must exist not-all-zero weights

$$w_j \in \{0, 1\} = \mathbb{F}_2 \quad \text{for } j \in \{1, \dots, n+1\}$$

such that

$$\sum_{j=1}^{n+1} w_j v_j = \mathbf{0}. \tag{3}$$

If we let $B = \{a_j \mid w_j = 1\}$, then

$$P = \prod_{b \in B} b = \prod_{j=1}^{n+1} a_j^{w_j}.$$

By (3), we know that

$$\sum_{j=1}^{n+1} e_i(a_j^{w_j}) \equiv 0 \pmod{2} \quad \forall i \in [n],$$

i.e., each prime exponent in the prime factorization of P is even, and therefore P is a perfect square. And since the weights $w_1, \dots, w_{n+1}$ are not all zero, B is a nonempty subset of A . $\square$

Solution 5. The number of times the die is rolled until it lands on each face at least once is a sum of n random variables $T_1, \dots, T_n$, where T_k is the number of times the die rolls before we see the k th new face.

So T_1 is always 1 (since we haven't seen any faces before it), T_2 is the number of rolls before we see a face other than the first, T_3 is the number of rolls before we see a face other than the first two, and so on.

⁶This is due to properties of exponentiation: since $(x^a y^b)^c = x^{ac} y^{bc}$, then (in reverse) $\prod_{i=1}^n p_i^{2r_i} = (\prod_{i=1}^n p_i^{r_i})^2$.

For a given T_k the probability of success (seeing a new face) is

$$\begin{aligned}
& 1 - \text{probability of failure} \\
&= 1 - \frac{k-1}{n} \\
& \quad (\text{we fail if we land on any of the } k-1 \text{ we've seen so far}) \\
&= \frac{n-(k-1)}{n}, \tag{4}
\end{aligned}$$

Suppose a trial succeeds with probability p . If X is “the number of independent runs of a trial until it succeeds,” then the expectation of X is (by definition)

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} \text{probability the } k\text{th trial succeeds} \cdot k.$$

The probability the 1st trial succeeds is p . The probability the second succeeds (and the first fails) is $1-p$. Therefore the probability that the k th trial succeeds (and all previous ones fail) is $(1-p)^{k-1}p$. So

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} (1-p)^{k-1}pk.$$

Let $x = (1-p)$. Then

$$\begin{aligned}
\mathbb{E}[X] &= (1-x) \sum_{k=1}^{\infty} x^{k-1}k \\
&= \sum_{k=1}^{\infty} x^{k-1}k - \sum_{k=1}^{\infty} x^k k \\
&= 1 + \sum_{k=2}^{\infty} x^{k-1}k - \sum_{k=1}^{\infty} x^k k \\
&= 1 + \sum_{k=1}^{\infty} x^k(k+1) - \sum_{k=1}^{\infty} x^k k \\
& \quad (\text{Incrementing } k \text{ in the summand requires} \\
& \quad \text{decrementing its start value to maintain equality}) \\
&= 1 + \sum_{k=1}^{\infty} (k+1)x^k - \sum_{k=1}^{\infty} kx^k = 1 + \sum_{k=1}^{\infty} x^k.
\end{aligned}$$

Now let

$$\begin{aligned}
S &= 1 + \sum_{k=1}^{\infty} x^k \iff Sx = \sum_{k=1}^{\infty} x^k \iff Sx = S - 1 \\
&\iff 1 = S(1-x) \iff S = \frac{1}{1-x}.
\end{aligned}$$

Granted, this only holds since $x < 1$. Now restore $x = (1 - p)$. Then

$$S = \frac{1}{1 - (1 - p)} = \frac{1}{p} = \mathbb{E}[X].$$

Therefore, by (4), $\mathbb{E}[T_k] = \frac{n}{n - (k - 1)}$, and the expected number of rolls of the die until each face lands at least once is

$$\sum_{k=1}^n \mathbb{E}[T_k] = \sum_{k=1}^n \frac{n}{n - (k - 1)} = n \left(\frac{1}{n} + \frac{1}{n - 1} + \cdots + \frac{1}{2} + \frac{1}{1} \right) = n \sum_{k=1}^n \frac{1}{k}.$$

I.e., the expected number of rolls of the die until it lands on each face at least once is nH_n , where H_n is the n th harmonic number.

Solution 6. Let's first answer "how many times is light $n \in [100]$ flipped?" Considering the example $75 = 5^2 3$, we see that it will be flipped by person 1, 3, 5, 15, 25, and 75: six times. That is, we notice that if switch n has k divisors, it is flipped exactly k times.

How many divisors does an arbitrary positive integer have? Well, if we consider its prime factorization,

$$n = \prod_{i=1}^{\ell} p_i^{e_i},$$

then we notice that to make a divisor of n we can choose between 0 and e_i "multiplicative copies" of p_i for each p_i independently. That is, a positive integer n with above prime factorization has exactly

$$k = \prod_{i=1}^{\ell} e_i + 1$$

divisors.

Now recall that the switches are originally all off, so a switch ends up on only if it is flipped an **odd number of times**. I.e., the switches which remain on are the ones with **an odd number of divisors**.

For a number to have an odd number of divisors, each e_i in its prime factorization must be **even**, because if just one e_j is odd, then $e_j + 1$ is even, and the entire product, k , becomes even.

And if every e_i is even, then n is a perfect square. I.e., the lights which remain on are those in

$$\{n^2 \mid n \leq 10\} = \{1, 4, 9, 16, 25, 36, 49, 64, 81, 100\}.$$

Solution 7.

Subsolution (Average number of loops). If there are n noodles, there are $2n$ noodle ends. After whichever end is chosen first, there are $2n - 1$ ends remaining

to choose from. The only way a loop forms is if the other end of the first noodle end is chosen, which there is only one of. So the probability that a loop forms is $\frac{1}{2n-1}$.

Alternatively, there are $\binom{2n}{2}$ possible ways to choose two noodle ends, n of which connect the ends of the same noodle. $\binom{2n}{2}$ simplifies to

$$\begin{aligned}\binom{2n}{2} &= \frac{2n!}{(2n-2)!2!} \\ &= \frac{\cancel{2}n(2n-1)(\cancel{2n-2})!}{\cancel{2}(\cancel{2n-2})!} \\ &= n(2n-1),\end{aligned}$$

so $n/\binom{2n}{2} = \frac{1}{2n-1}$.

Let L_n be a random variable that is 1 if a loop forms out of n noodles and 0 otherwise. Each time we connect two noodle ends—whether two different noodles connect or a loop forms—the number of effective noodles decreases by one each time. Then the average number of noodles that form if there are originally 100 is

$$\begin{aligned}\mathbb{E}[L_{100} + L_{99} + \cdots + L_1] &= \sum_{n=1}^{100} \mathbb{E}[L_n] && \text{(linearity of expectation)} \\ &= \sum_{n=1}^{100} \frac{1}{(2n-1)} \\ &= 1 + \frac{1}{3} + \frac{1}{5} + \cdots + \frac{1}{199} \\ &\approx 3.2843.\end{aligned}$$

For general n , $1 + \frac{1}{3} + \cdots + \frac{1}{2n-1} = H_{2n} - \frac{1}{2}H_n$, where H_n is the n th harmonic number.

Subsolution (Probability one loop forms). The only way a single loop forms is if a loop never forms until there is just one noodle left (at which point a loop must form). The probability a loop doesn't form out of n noodles is $1 - \frac{1}{2n-1} = \frac{2n-2}{2n-1}$. Then the desired probability is

$$\begin{aligned}\prod_{n=2}^{100} \frac{2n-2}{2n-1} &= \prod_{n=1}^{99} \frac{2(n+1)-2}{2(n+1)-1} \\ &= \prod_{n=1}^{99} \frac{2n}{2n+1} \\ &= \frac{2}{3} \cdot \frac{4}{5} \cdot \frac{6}{7} \cdot \cdots \cdot \frac{198}{199} \\ &\approx 0.088734.\end{aligned}$$

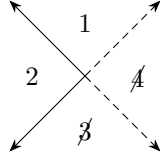
Solution 8. You can always draw $n - 1$ lines in different directions so no two are parallel and at different positions so that none intersect more than one line at the same point. By doing so, the n th line will cross all previous $n - 1$ lines. As it crosses each one, a new region is created. Also, by passing through the entire plane, the n th line creates another region. That is, the n th line will produce $n - 1 + 1 = n$ new regions. So we get the recurrence

$$\begin{aligned} L_0 &= 1; \\ L_n &= L_{n-1} + n \quad \forall n \geq 1. \end{aligned}$$

Unfolding this we see

$$\begin{aligned} L_1 &= L_0 + 1 = 2, \\ L_2 &= L_1 + 2 = 4, \\ L_3 &= L_2 + 3 = 7, \\ L_n &= 1 + \sum_{i=1}^n i \\ &= \frac{1}{2}n^2 + \frac{1}{2}n + 1. \end{aligned} \quad \left(\sum_{i=1}^n i = \frac{n(n+1)}{2} \right) \quad (5)$$

Solution 9. A zig is just like two lines, except *each one creates two less regions*.

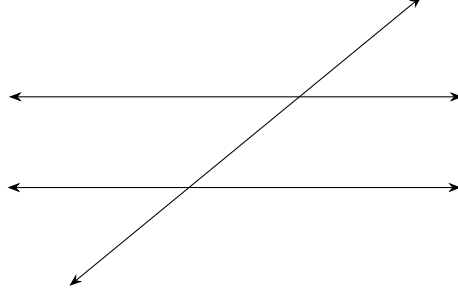


So

$$\begin{aligned} Z_n &= L_{2n} - 2n \\ &= \frac{1}{2}(2n)^2 + \frac{1}{2}(2n) + 1 - 2n \quad \text{from (5)} \\ &= 2n^2 - n + 1. \end{aligned}$$

Solution 10. We start by considering an infinite zig-zag (IZZ): three infinite

lines but two are parallel.



To find the maximum number of regions n IZZes segment 2D space into, IZ_n , we find the number of regions each line contributes consecutively. Supposing the third line is parallel to one of the earlier two,

$$\begin{aligned}
 IZ_n &= IZ_{n-1} + 3(n-1) + 1 && \text{from (1)} \\
 &\quad + 3(n-1) + 2 && \text{from (2)} \\
 &\quad + 3(n-1) + 3 - 1 && \text{from (3)} \\
 &= IZ_{n-1} + 9(n-1) + 5 \\
 &= 9n - 4 + IZ_{n-1}.
 \end{aligned} \tag{6}$$

- (1) The first line crosses $3(n-1)$ lines, adding $3(n-1) + 1$ new regions.
- (2) The second line crosses $3(n-1) + 1$ lines, adding $3(n-1) + 2$ new regions.
- (3) The third line can cross all $3(n-1) + 2$ lines *except one*—whichever of the earlier two it is parallel to, creating $3(n-1) + 3 - 1$ new regions.

Unfolding recurrence (6), we see that for $n \geq 1$

$$\begin{aligned}
 IZ_n &= 9(n) - 4 + 9(n-1) - 4 + \cdots + 9(1) - 4 + IZ_0 \\
 &= 9 \left(\sum_{i=1}^n i \right) - 4n + 1 && (IZ_0 = 1) \\
 &= \frac{9}{2}n^2 + \frac{9}{2}n - \frac{8}{2}n + 1 && \left(\sum_{i=1}^n i = \frac{n(n+1)}{2} \right) \\
 &= \frac{9}{2}n^2 + \frac{1}{2}n + 1.
 \end{aligned} \tag{7}$$

So $IZ_2 = \frac{9}{2}2^2 + \frac{1}{2}2 + 1 = 18 + 1 + 1 = 20$, as shown in Figure 1.

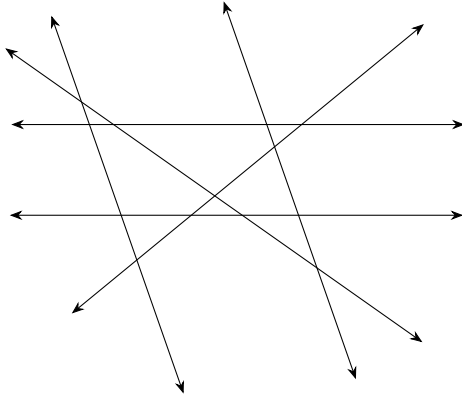
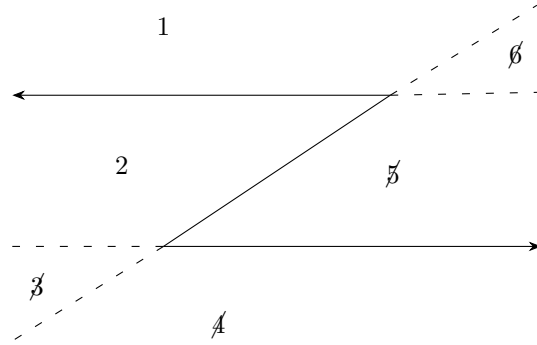


Figure 1: Two IZZes segment the plane into 20 regions

Now back to the original zig-zag to find ZZ_n . The zig-zag is just like an IZZ, except *each one creates four less regions*.



So

$$\begin{aligned}
 ZZ_n &= IZ_n - 4n \\
 &= \frac{9}{2}n^2 + \frac{1}{2}n + 1 - \frac{8}{2}n && \text{from (7)} \\
 &= \frac{9}{2}n^2 - \frac{7}{2}n + 1.
 \end{aligned}$$